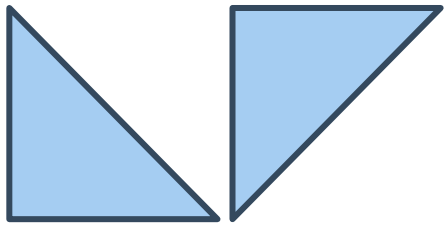
Translations

1 State the type of transformattion that has occured on the following shapes:

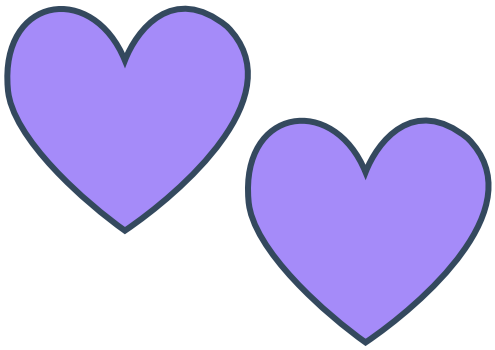
a



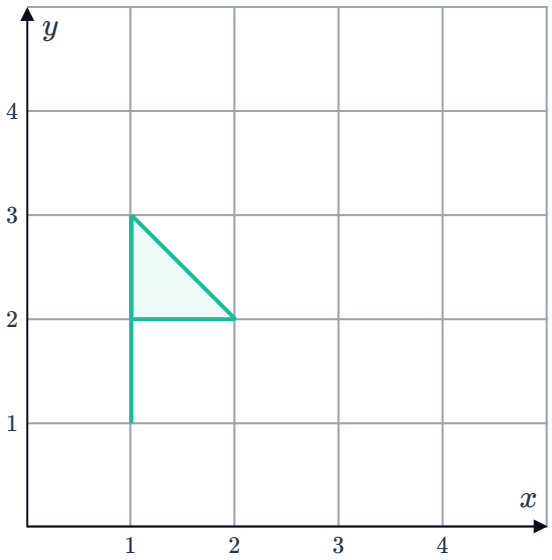
b



c



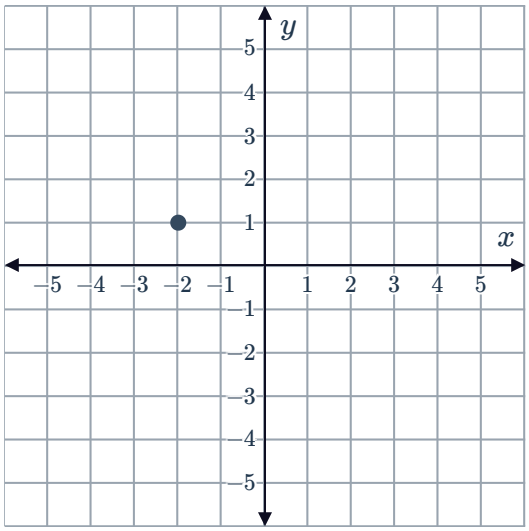
2 Translate the given flag 1 unit up and 2 units to the right on the number plane:



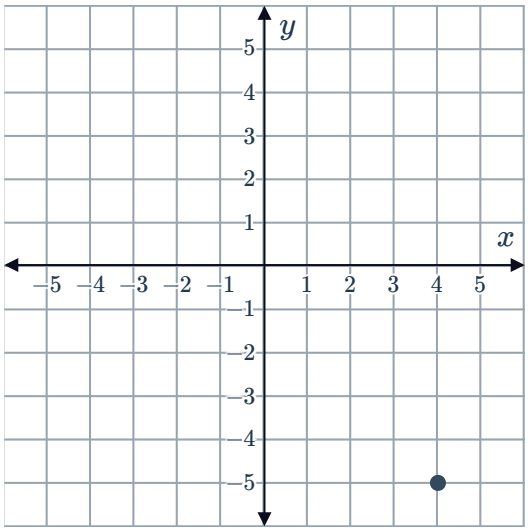
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3 For each of the following points, plot the translated point using the given instruction:

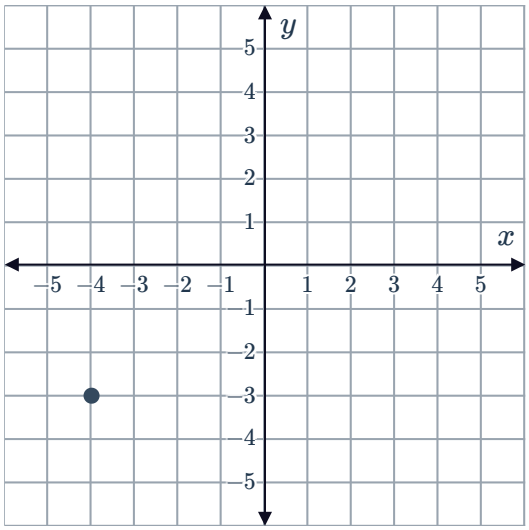
a Translate the point 3 units to the right.



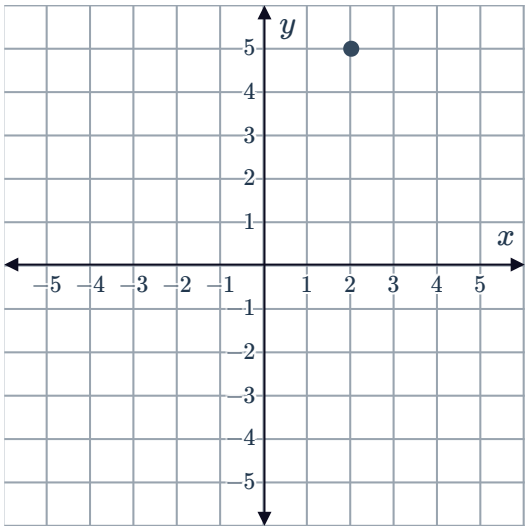
b Translate the point 4 units to the left.



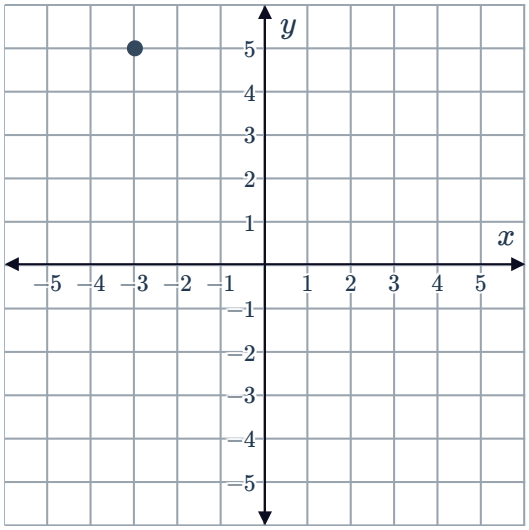
c Translate the point 7 units up.



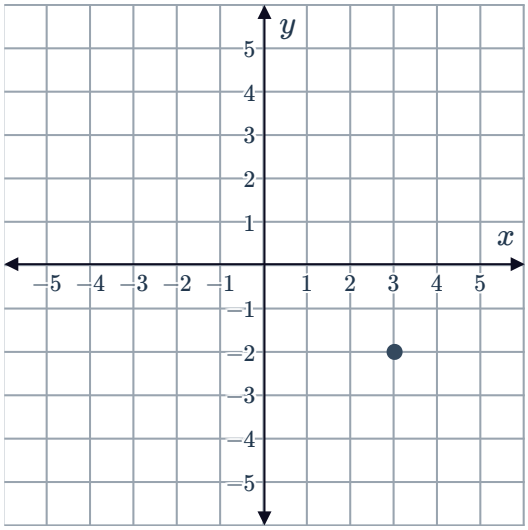
d Translate the point 4 units down.



e Translate the point 2 units to the right and 4 units down.



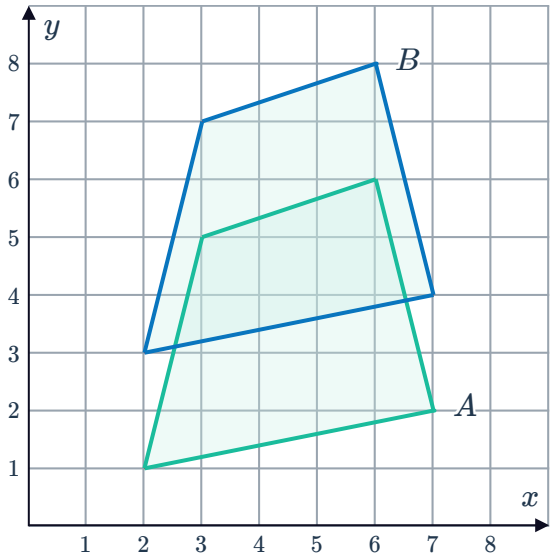
f Translate the point 5 units to the left and 6 units up.



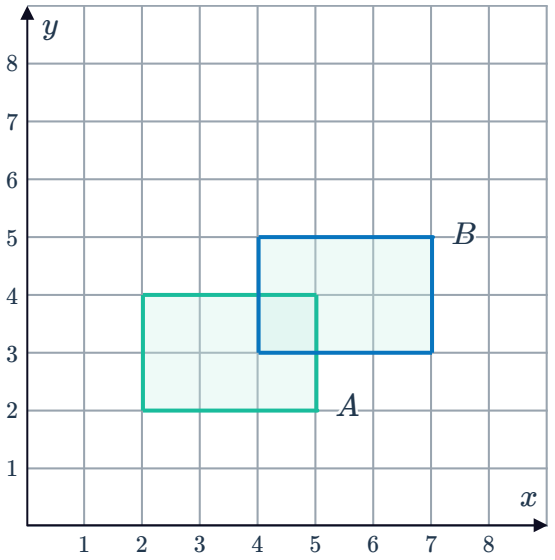
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4 For each of the following, describe the translation from figure A to figure B:

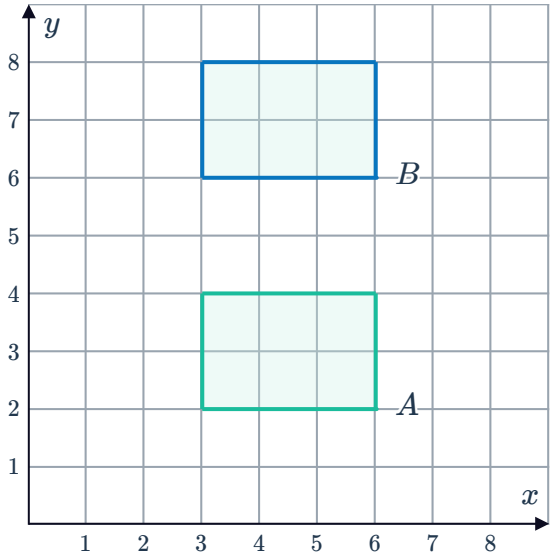
a



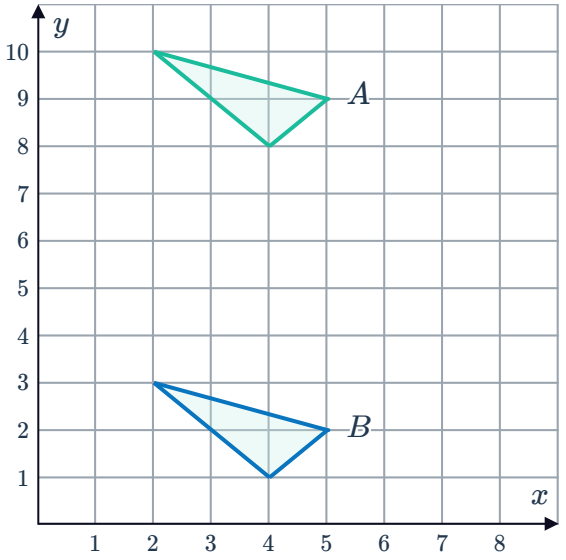
b



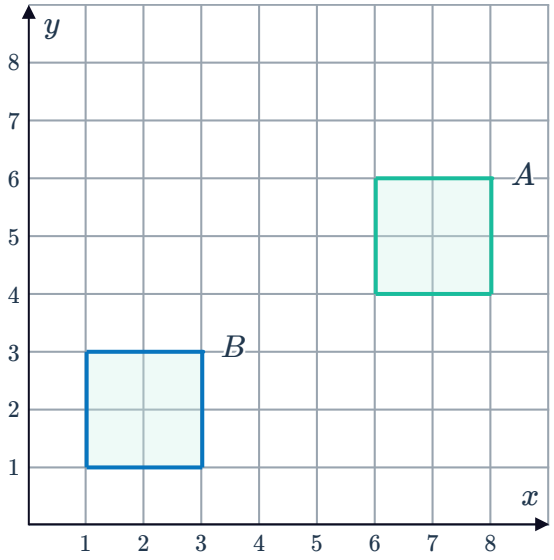
c



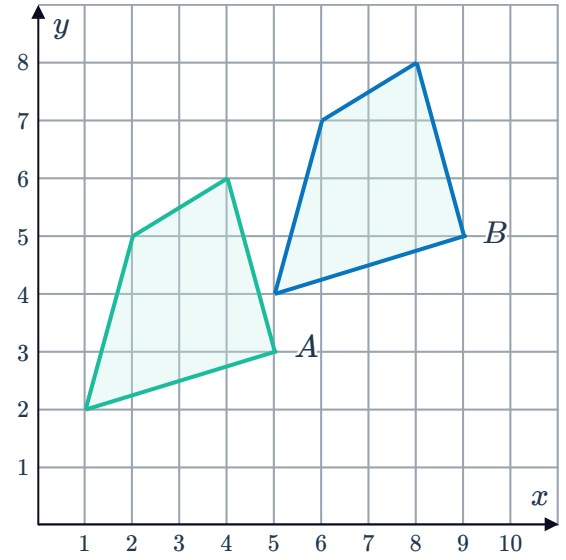
d



e

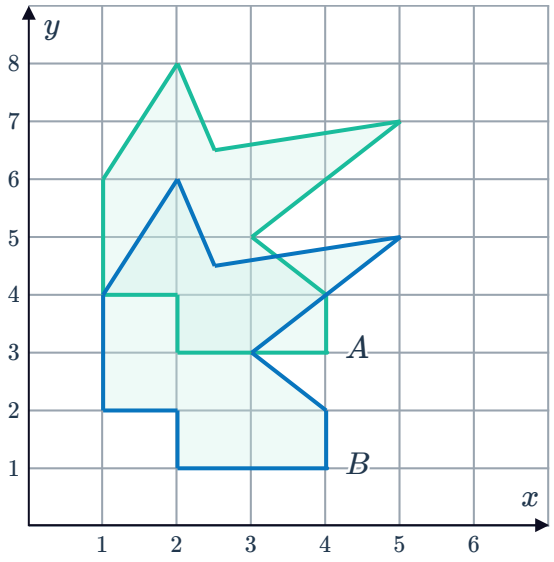


f

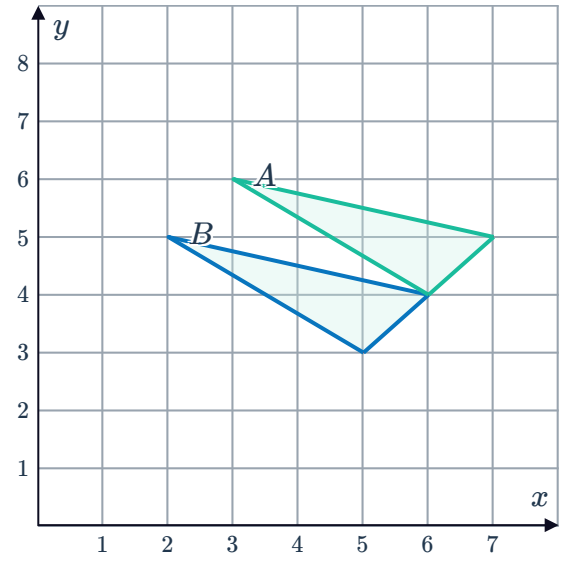


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g

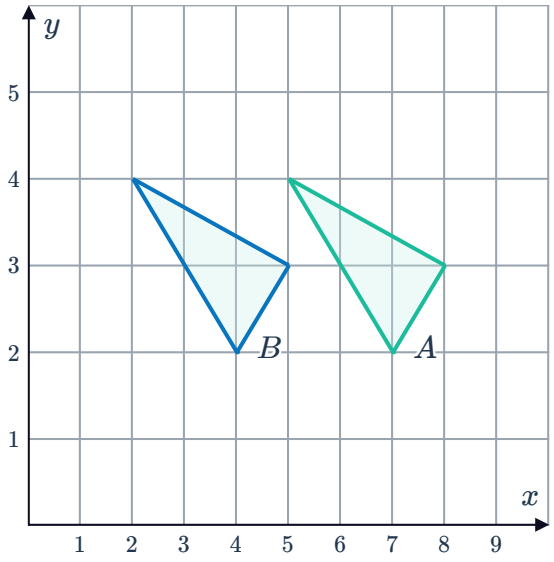


h

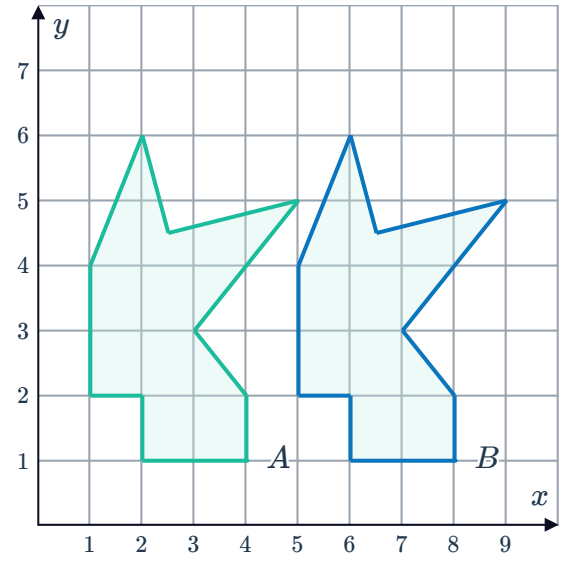


5 For each of the following, describe the translation from figure B to figure A:

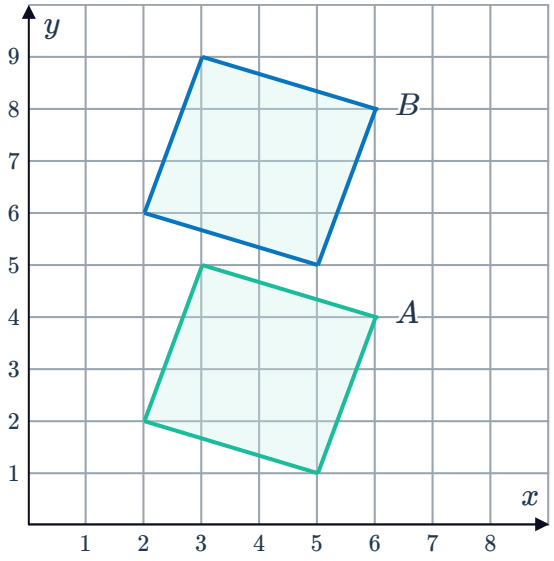
a



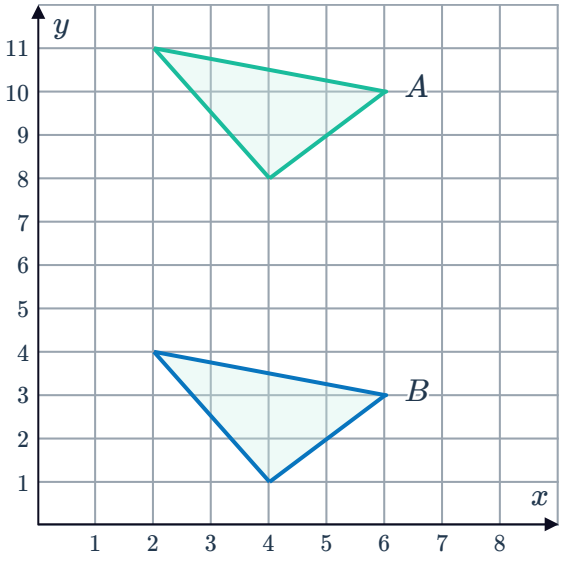
b



c

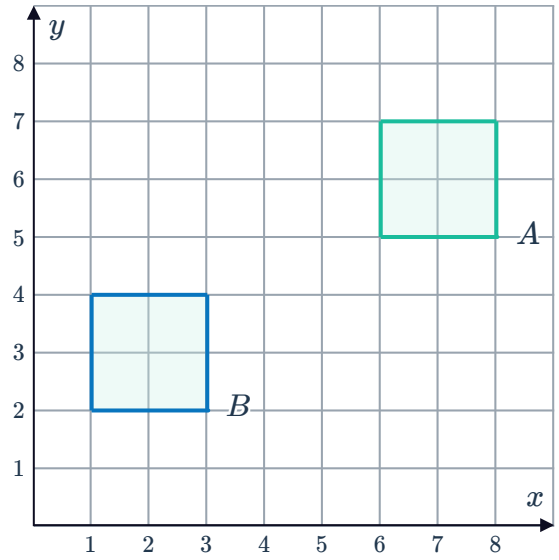


d

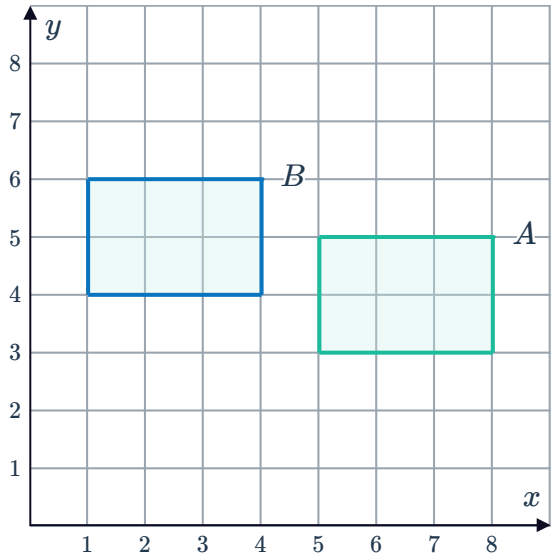


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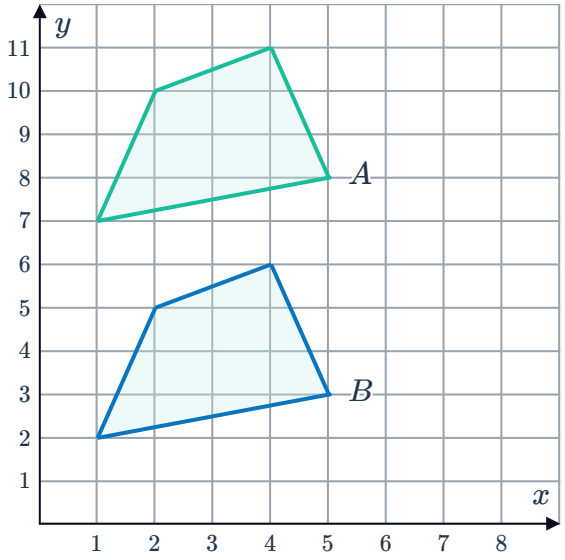
e



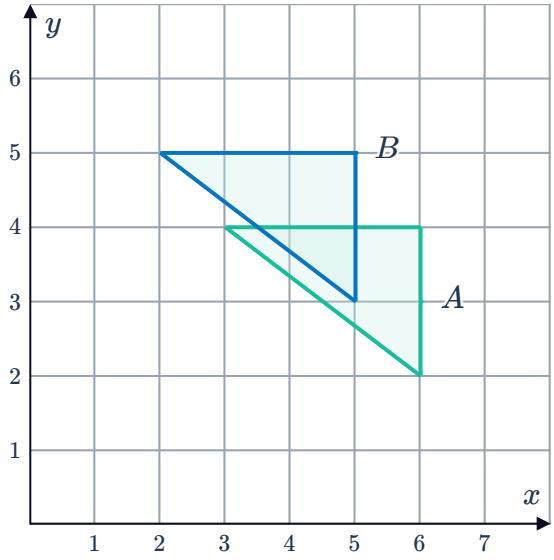
f



g

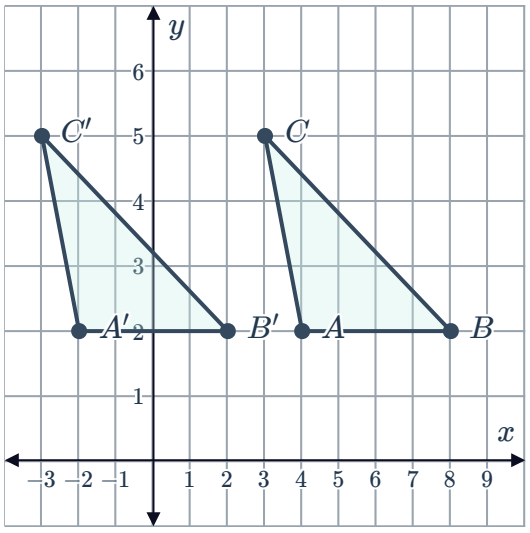


h

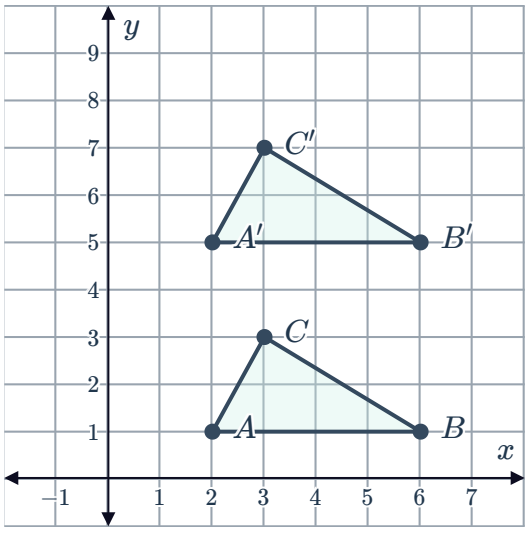


6 For each of the following, describe the translation of triangle ABC to triangle $A'B'C'$:

a

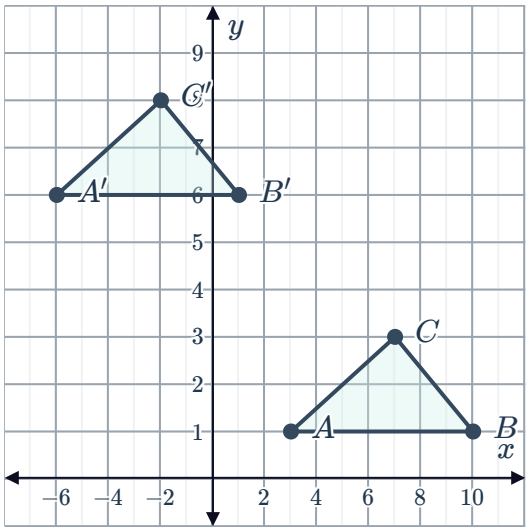


b



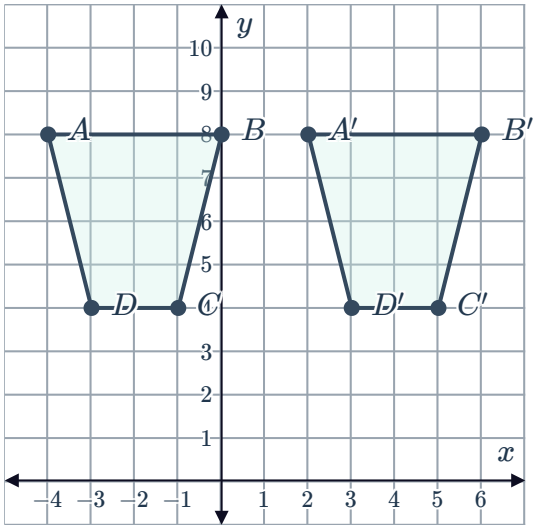
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c

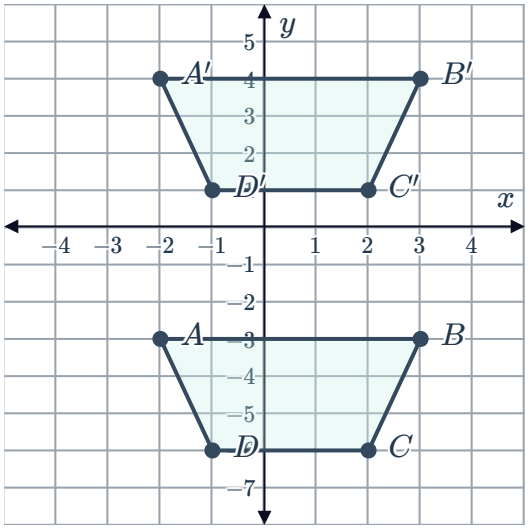


7 For each of the following, describe the translation of the trapezium $ABCD$ to the trapezium $A'B'C'D'$:

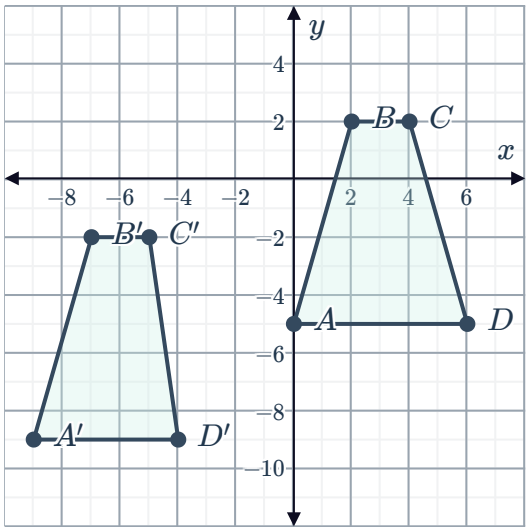
a



b



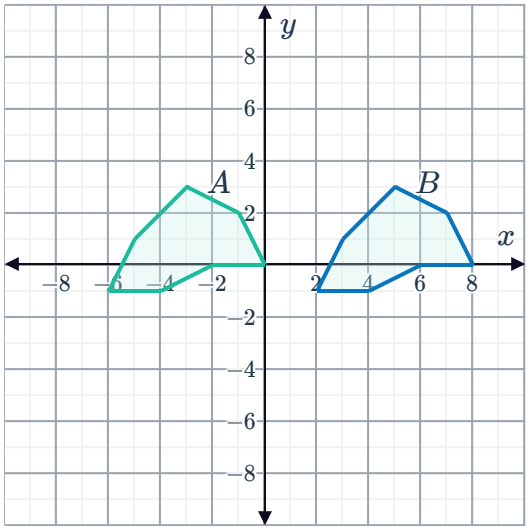
c



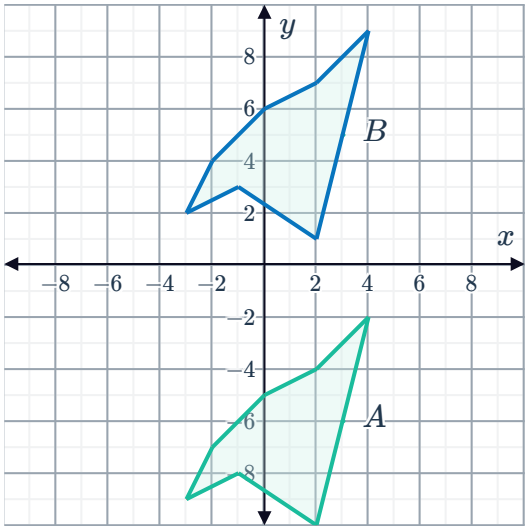
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8 For each of the following, describe the translation of the polygon A to the polygon B:

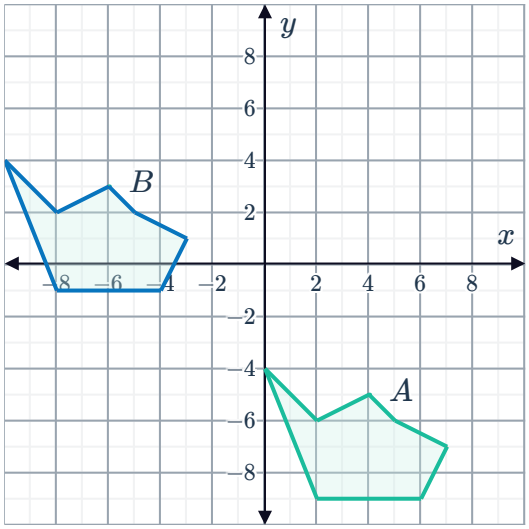
a



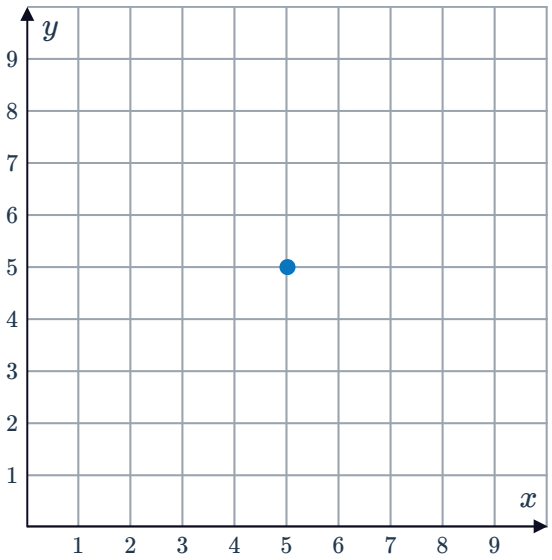
b



c

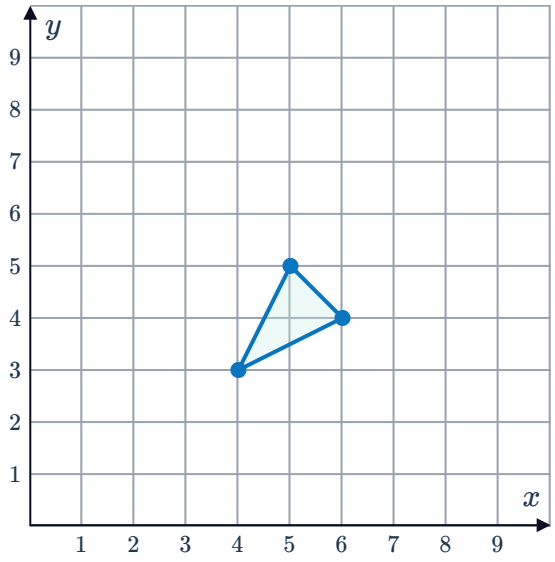


9 A point has been translated 2 units left. The translated point is shown on the right. Plot the starting position of this point before it was translated.

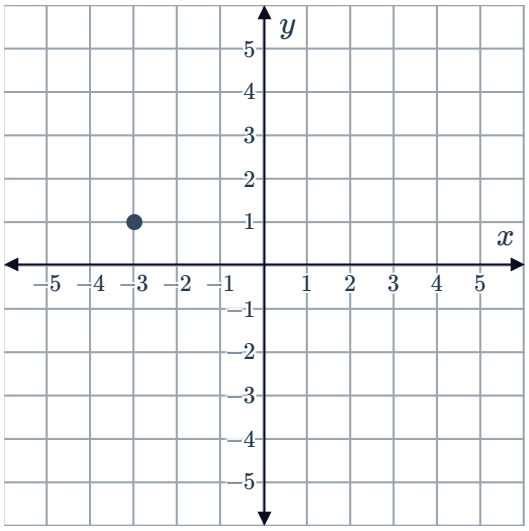


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10 A triangle has been translated 3 units left. The translated triangle is shown on the right. Plot the starting position of the triangle before it was translated.



11 The point on the number plane is translated 4 units up. Write down the coordinates of its new position.



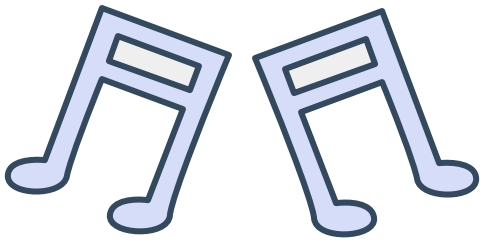
12 Point A is translated 3 units down and 4 units to the right, where it sits at point $(5, -2)$. Write down the coordinate of the original point A.

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Reflections

13 State the type of transformation that has occured on the following shapes:

a



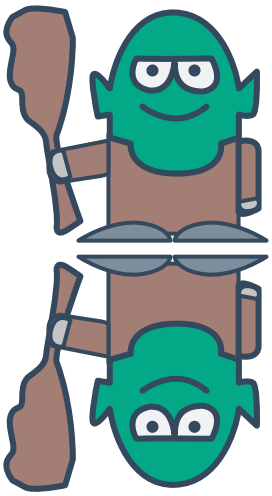
b



c



d



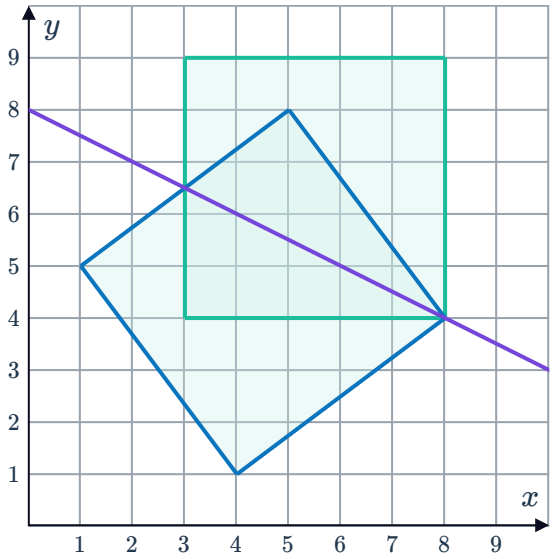
14 True or false:

A shape is reflected across a line. If the reflected shape is then reflected across the same line again, it will return to the original shape.

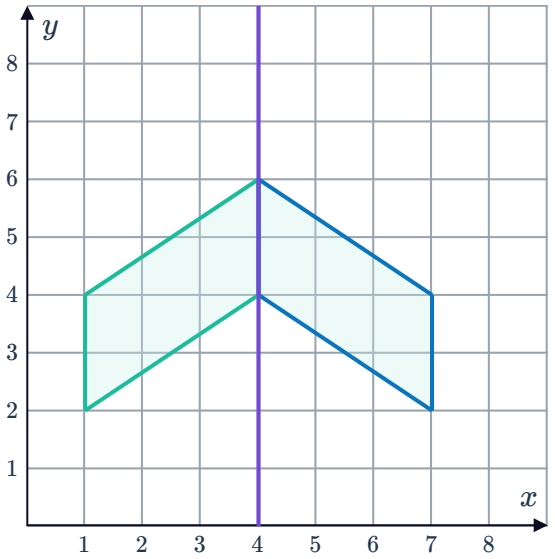
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15 Determine whether the following diagrams show a reflection across the given line:

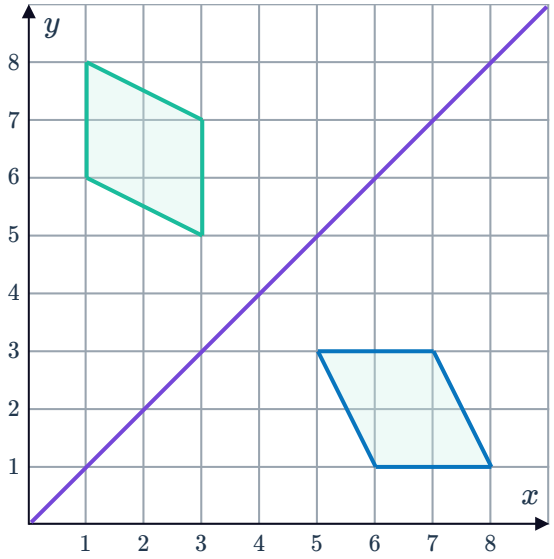
a



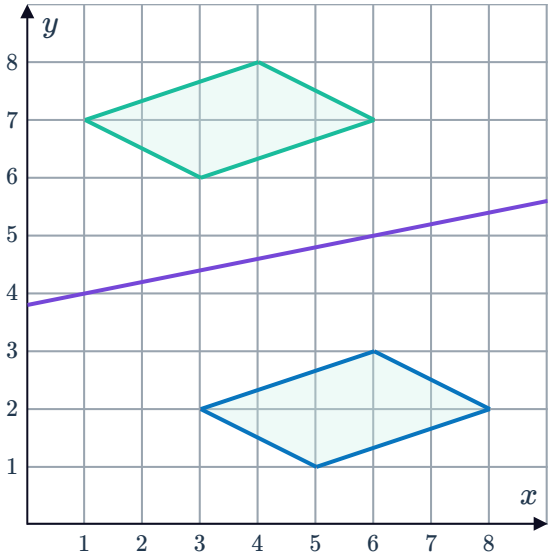
b



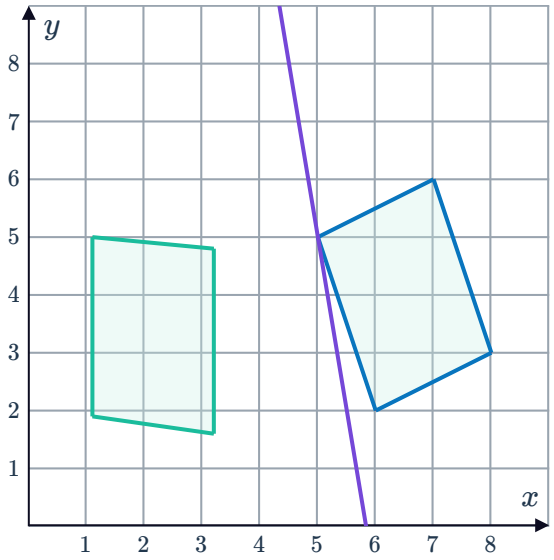
c



d



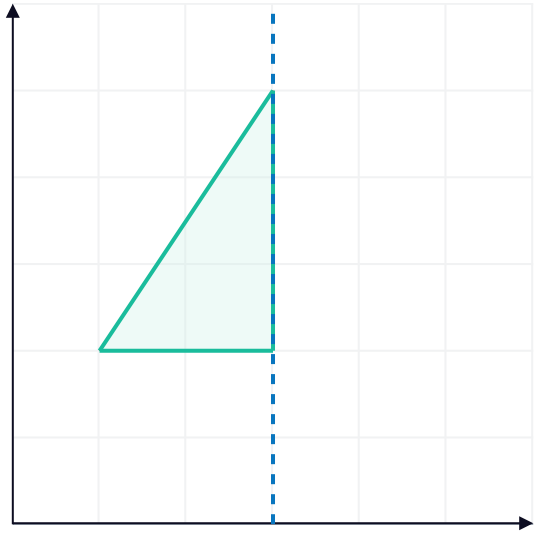
e



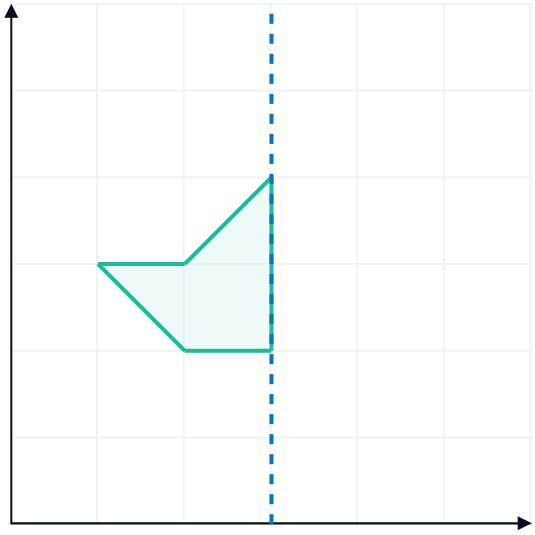
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16 Reflect the following shapes about the given line:

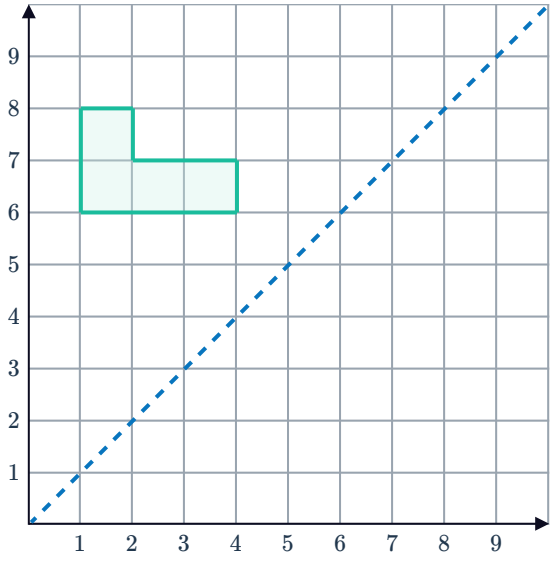
a



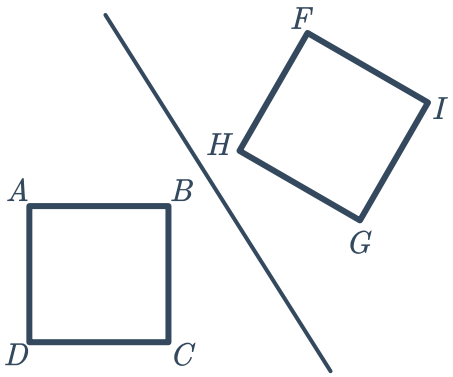
b



c



17 The square below is reflected about the given line. Which point would correspond to point A?



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18 For each of the following:

- Plot the point A on a number plane.
- Plot point A', a reflection of point A across the indicated axis.

a $A\,(8, 5)$; x-axis

b $A\,(-10, -6)$; x-axis

c $A\,(-5, -3)$; y-axis

d $A\,(8, -6)$; y-axis

19 Consider the point $A\,(5, 6)$.

a Plot point A on a number plane.

b If point A is reflected across the y-axis to produce point B, what will be the coordinates of point B?

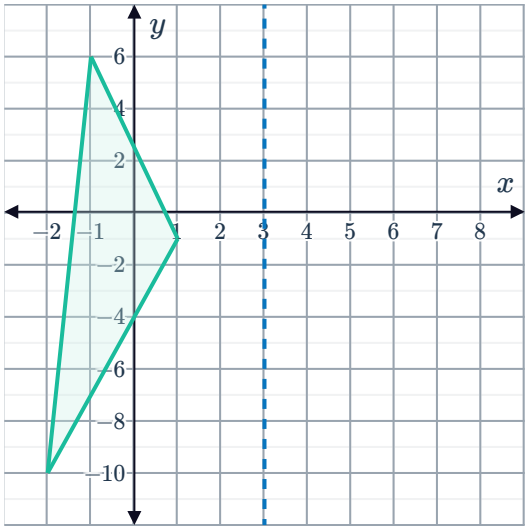
c What distance would the original point A need to be translated to overlap point B?

20 Consider the line segment AB, where the endpoints are $A\,(-9, -10)$ and $B\,(4, 3)$.

a Plot the line segment AB on a number plane.

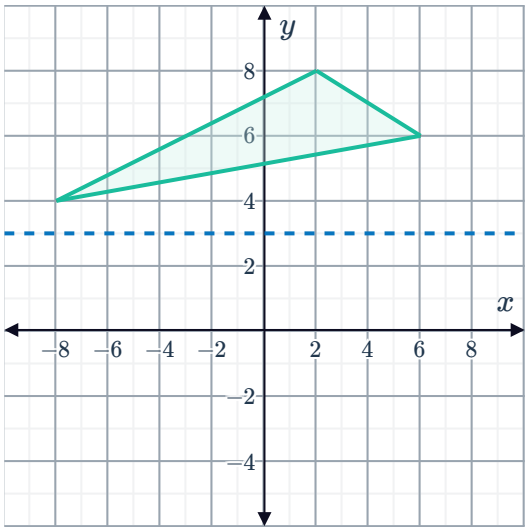
b Plot the reflection of the line segment AB across the x-axis.

21 Plot the reflection of the given triangle across the line $x = 3$:

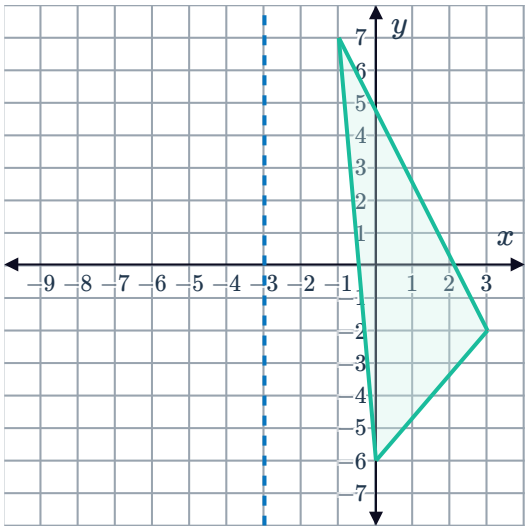


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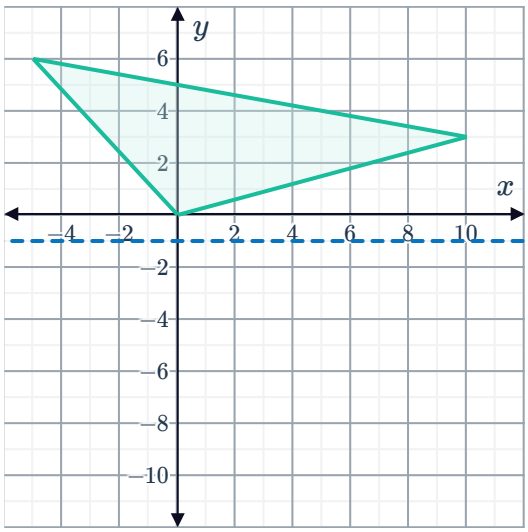
22 Plot the reflection of the given triangle across the line $y = 3$:



23 Plot the reflection of the given triangle across the line $x = -3$:



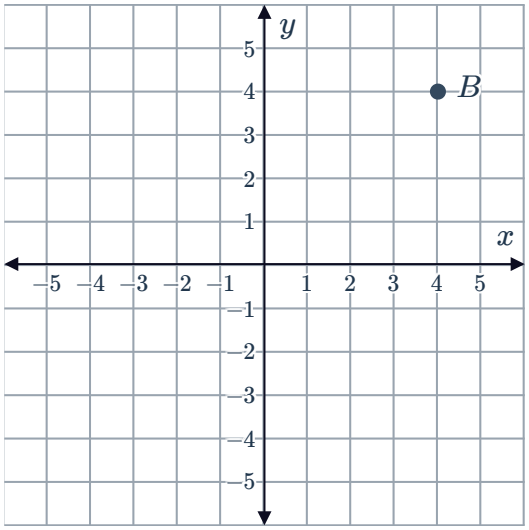
24 Plot the reflection of the given triangle across the line $y = -1$:



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Rotations

25 Find the coordinates of point B when it is rotated 90° clockwise about the origin.



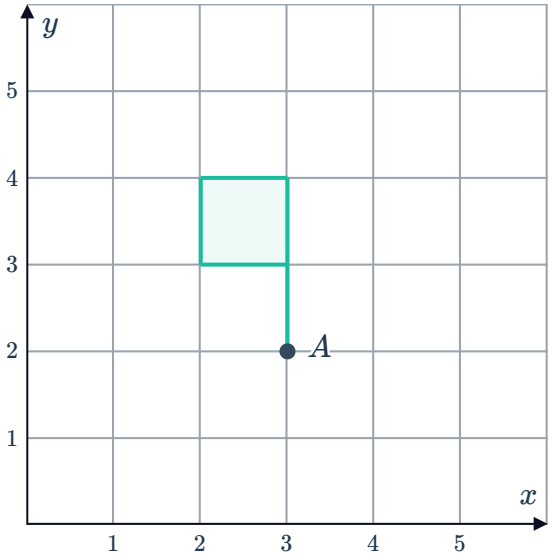
26 Consider the flag on the number plane shown:

Draw the flag after it has been rotated:

a 90° about point A.

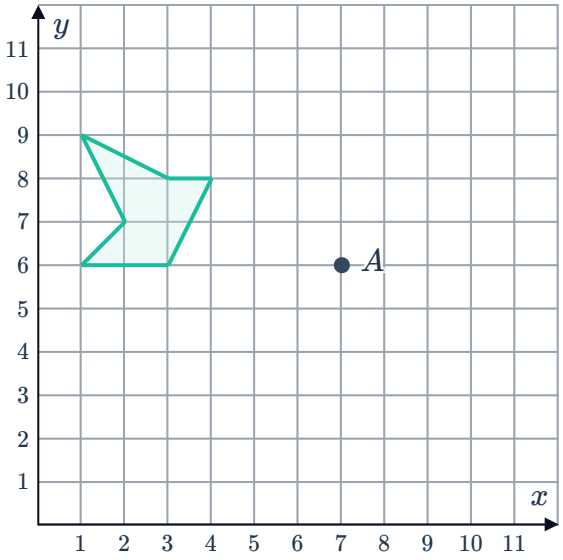
b 180° about point A.

c 270° about point A.



27 Consider the following shape:

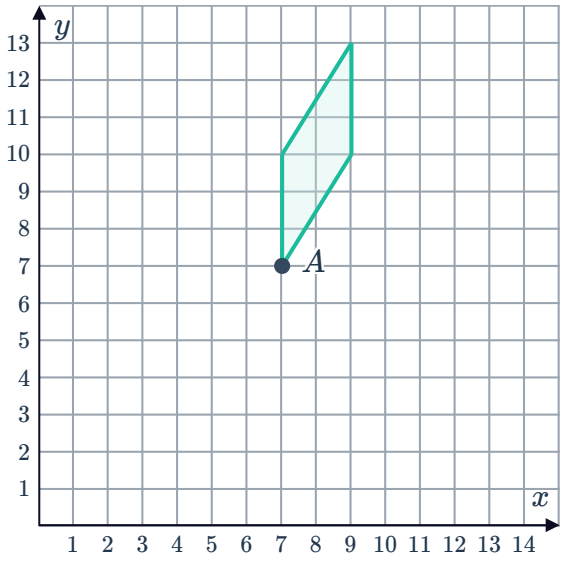
Draw the shape after it has been rotated by 180° clockwise about point A.



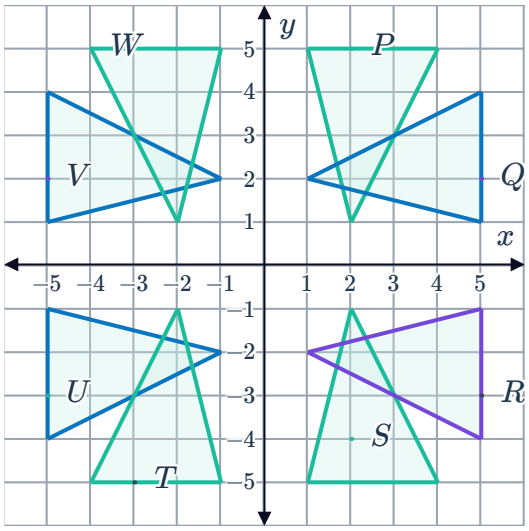
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28 Consider the shape shown:

Draw the shape after it has been 135° anticlockwise rotated about point A.



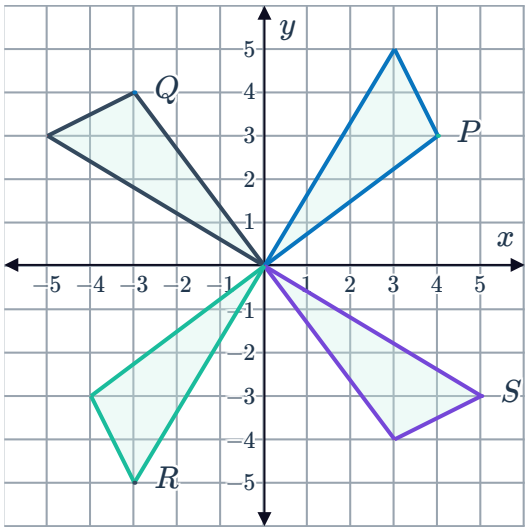
29 Which shape corresponds to a rotation of triangle R by 270° clockwise about the origin?



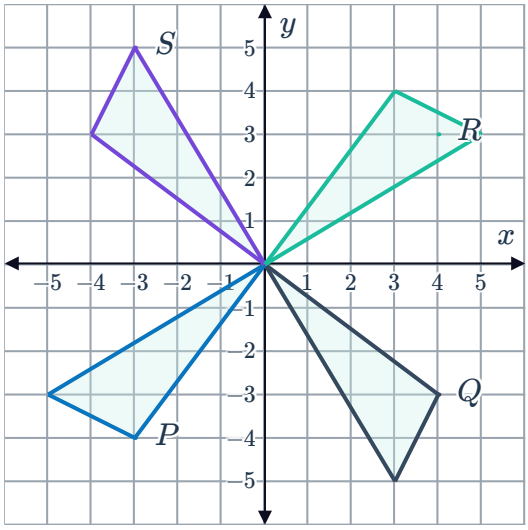
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30 Suppose all four triangles in the image below are rotated 270° clockwise about the origin.

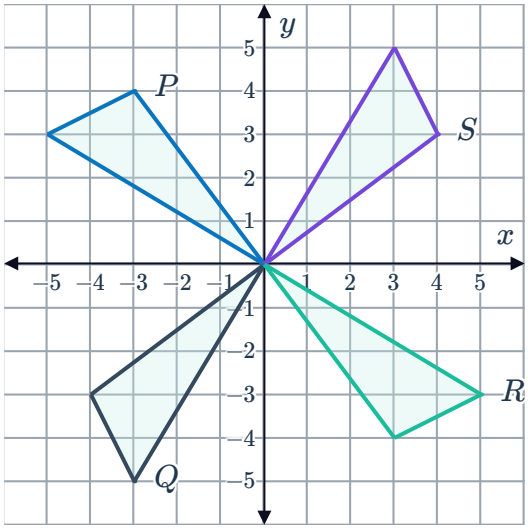
Which image would correspond to the result of this transformation?



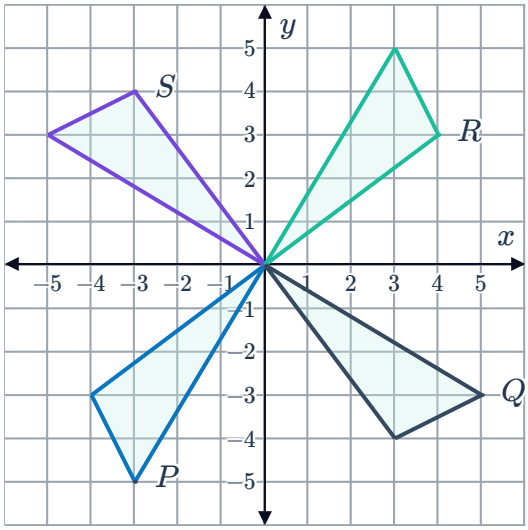
A



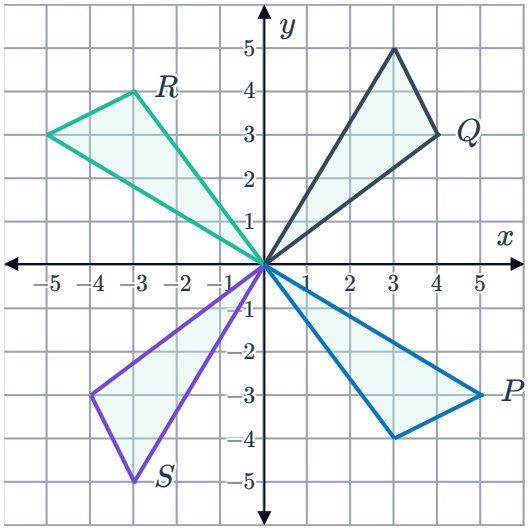
B



C



D



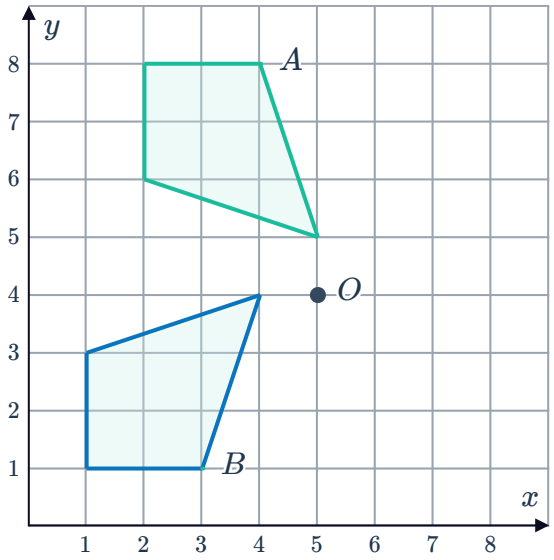
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31 For each of the following:

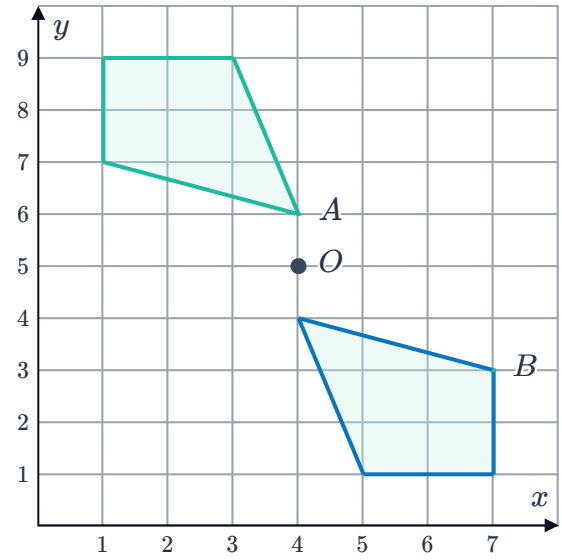
i Find the clockwise angle of rotation of shape A to shape B about point O.

ii Find the anticlockwise angle of rotation of shape A to shape B about point O.

a



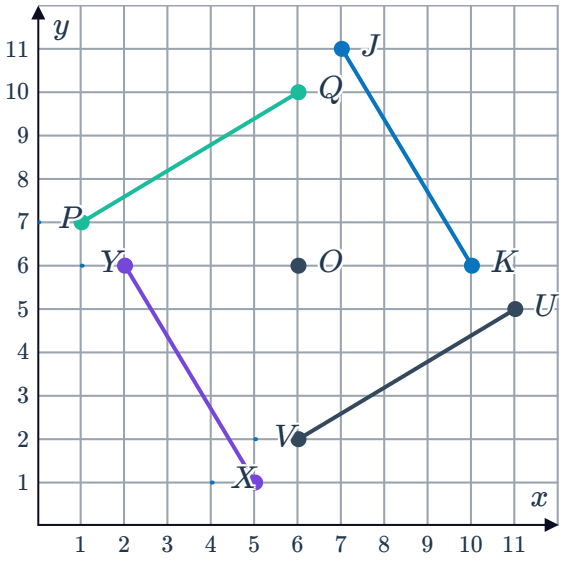
b



32 Consider the following image:

a Which segment represents a clockwise rotation of segment PQ by 90° about point O?

b Which segment represents an anticlockwise rotation of segment XY by 270° about point O?



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